

VOLATILITY-SIGNAL PORTFOLIO CONSTRUCTION

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1. Overview of the Methodological Framework

This study constructs a systematic quantitative equity model for the Vietnamese stock market based on historical volatility and related risk indicators. The objective is to evaluate whether lagged stock-level risk information can separate future return performance across the VN100 universe. Unlike a directional market-timing model, the proposed framework is a cross-sectional ranking strategy. At each monthly formation date, stocks are sorted within sectors into three portfolios from low risk to high risk. The model then compares the subsequent performance of P1, P2, and P3.

The strategy is motivated by the empirical literature on low-volatility and low-beta equity effects. Prior studies show that high-volatility or high-beta stocks do not always earn proportionately higher subsequent returns, creating a potential low-risk anomaly in cross-sectional equity returns. The present framework adapts that idea to the Vietnamese equity market by testing three simple risk indicators: realized standard deviation, maximum drawdown, and market beta. The use of simple transparent indicators is intentional. It allows the backtest to focus on whether historical risk measures contain useful ranking information before adding more complex machine-learning or multi-factor extensions.

The model is long-only throughout. No short portfolio is implemented, although the analysis reports the P1 minus P3 spread as a diagnostic measure of signal strength. This design is consistent with the practical constraints of the Vietnamese equity market, where short selling is limited and difficult to implement at scale. Monthly rebalancing is selected to balance signal responsiveness against turnover and transaction costs.

The methodology follows the same broad architecture as the value and momentum framework used in the reference implementation: data cleaning, signal construction, within-sector portfolio assignment, sector-replicated market-capitalization weighting, turnover and trading-cost computation, and benchmark comparison. The reference file uses within-sector tertile sorting and sector-replicated market-capitalization weighting to reduce sector concentration and preserve investability. The volatility model retains these portfolio

construction principles while replacing the value and momentum signals with risk-based signals.

2. Data

2.1 Price Data

Daily adjusted closing prices and trading volumes are obtained for 100 stocks included in VN100, covering the period from January 2015 to May 2026. All price series are adjusted for corporate actions including stock splits, reverse splits, and rights issues.

The cleaned daily price matrix is converted into wide format with dates as rows and tickers as columns. The model uses daily returns defined from this matrix. Monthly rebalance dates are identified as the last available trading date in each calendar month. Holding-period returns are calculated directly from the portfolio formation date to the next monthly formation date, using the latest available price at those dates.

2.3 Market Capitalisation Data

Market capitalization data are used for portfolio weighting. The default market capitalization input contains ticker, date, and market capitalization. obtained from FiinProX for all stocks in the universe, covering the same sample period. Market capitalisation is defined as the product of the closing share price and the number of shares outstanding on the corresponding date. Market capitalization is forward-filled to each monthly formation date using the most recent observation available on or before that date. This procedure avoids the use of future market capitalization information when constructing portfolio weights.

3. Signal Construction

3.1 Daily Return and Volatility Window

Let $P_{i,t}$ denote the adjusted closing price of stock i on trading day t . The daily simple return is defined as:

$$r_{i,t} = \frac{P_{i,t-1}}{P_{i,t}} - 1 \quad (3.1)$$

The benchmark return is defined analogously using the VNINDEX closing price:

$$r_{m,t} = \frac{P_{m,t-1}}{P_{m,t}} - 1 \quad (3.2)$$

The use of recent historical volatility as a stock-ranking signal is motivated by a broad empirical literature showing that volatility contains information relevant to subsequent returns and portfolio risk. Ang, Hodrick, Xing, and Zhang (2006) document a negative cross-sectional relation between idiosyncratic volatility and future average returns in the U.S. equity market, showing that stocks with high recent idiosyncratic volatility tend to earn lower subsequent returns. Ang, Hodrick, Xing, and Zhang (2009) extend this evidence internationally and find that the negative relation between high idiosyncratic volatility and low future returns is present across a broad sample of developed equity markets. These findings support the basic empirical premise of sorting stocks by recent volatility and comparing the future performance of low-risk and high-risk portfolios.

Related evidence also supports the use of historical volatility in portfolio construction rather than treating volatility only as a descriptive risk measure. Fleming, Kirby, and Ostdiek (2001) show that volatility timing can have economic value in short-horizon asset allocation because conditional variances and covariances are generally more precisely estimated than conditional expected returns. Moreira and Muir (2017) further show that volatility-managed portfolios, which reduce exposure when recent realized volatility is high, can improve Sharpe ratios because expected returns do not rise proportionally with volatility. These studies justify using recent volatility not only to describe risk, but also to guide portfolio formation and performance testing.

At the same time, the volatility-return relation is sensitive to empirical design. Bali and Cakici (2008) show that the significance and even existence of the idiosyncratic volatility effect depend on choices such as estimation frequency, portfolio weighting method, breakpoint construction, and liquidity screens. This evidence is directly relevant to the present methodology because the Vietnamese equity market contains non-synchronous trading, occasional missing observations, and material differences in market capitalization across stocks. Therefore, the volatility signal must be constructed using a sufficiently stable estimation window and a minimum data-coverage rule.

The baseline volatility signals are estimated over a 60-trading-day rolling window, subject to a minimum requirement of 45 valid observations. The 60-day window is motivated by prior empirical evidence on volatility-based return prediction. Ang, Hodrick, Xing, and Zhang use recent daily returns to estimate stock-level volatility and show that high-volatility stocks tend to earn lower future returns. Their international evidence further tests volatility formation windows of three, six, and twelve months and finds that the negative volatility-return relation remains statistically significant. In particular, the three-month formation period performs strongly because it provides a more accurate volatility estimate than a shorter one-month window while still preserving recent information. Since 60 trading days correspond approximately to three trading months, the present methodology adopts a 60-day window as the baseline horizon for estimating standard deviation, maximum drawdown, and beta.

The choice of 60 trading days also fits the monthly rebalancing design of the strategy. A one-month volatility window would be more reactive, but it would be more exposed to short-term noise, price jumps, and temporary trading frictions. A much longer window, such as six or twelve months, would produce more stable estimates, but it would react more slowly to changes in risk conditions. The 60-day window therefore represents a middle-ground specification. It is long enough to improve the reliability of the risk estimate, but short enough to capture recent changes in stock-level volatility before the next monthly holding period.

The minimum-observation rule is imposed to ensure that the volatility estimate is not based on insufficient data. This follows the empirical practice in idiosyncratic-volatility studies of requiring a minimum number of valid daily returns before estimating stock-level volatility. For example, Bozhkov et al. (2020) require at least 15 non-zero daily returns in the preceding month and discard zero-return days to reduce the influence of infrequent trading. Similarly, Hou, Xue, and Zhang require at least 10 daily returns to compute IVOL and show that their results are robust when the threshold is increased to 15 daily returns. In anomaly-replication work, Hou, Xue, and Zhang also require a minimum of 15 daily returns when estimating previous-month idiosyncratic volatility for monthly portfolio sorts. Applying the same principle to a 60-trading-day window, the present methodology requires 45 valid observations, equivalent to 75% data coverage, so that each volatility signal is based on a sufficiently dense return history. The 45-observation threshold is particularly important in the Vietnamese equity market, where missing observations can arise from trading suspensions, non-trading days, stale prices, or incomplete data records. Requiring all 60 observations would be too restrictive and could unnecessarily exclude otherwise investable VN100 stocks. Allowing substantially fewer observations would increase coverage but would weaken the reliability of the volatility, drawdown, and beta estimates. The 45-out-of-60 rule therefore balances signal reliability with practical data availability.

3.2 Standard Deviation Signal

The first risk signal is the annualized standard deviation of daily stock returns over the trailing 60-trading-day window. It measures total realized volatility at the individual stock level

$$x_{i,t}^{STD} = \sqrt{252} \times \text{std}(r_{i,t-L+1}, \dots, r_{i,t}) \quad (3.2)$$

A lower standard deviation indicates lower realized risk. Therefore, stocks with lower $x_{i,t}^{STD}$ are assigned toward P1.

3.3 Maximum Drawdown Signal

The second risk signal is the maximum peak-to-trough loss over the trailing 60-trading-day price window. For each stock, the model identifies the largest decline from a previous local peak to a subsequent trough within the rolling window.

$$x_{i,t}^{MDD} = \max_{t-L+1 \leq a < b \leq t} \left[1 - \frac{P_{i,b}}{P_{i,a}} \right] \quad (3.3)$$

The drawdown value is expressed as a positive number. For example, a value of 0.20 represents a 20 percent maximum drawdown during the window. A lower maximum drawdown indicates lower recent downside risk and be assigned to P1.

3.4 Beta Signal

The third risk signal is the stock's rolling beta relative to VNINDEX. Beta measures the stock's sensitivity to market movements over the trailing 60-trading-day window.

$$x_{i,t}^{BETA} = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)} \quad (3.4)$$

where the covariance and variance are estimated using overlapping valid stock and VNINDEX daily returns inside the trailing window. A lower beta indicates lower market sensitivity and receives a better rank. Because beta requires both stock return data and VNINDEX return data, the first valid beta signal may appear later than the first valid standard deviation or maximum drawdown signal if benchmark data are missing, sparse, or misaligned.

3.5 Signal Interpretation

The three signals are tested separately. They are not combined into a composite score in the baseline implementation. The interpretation of the P1 portfolio is therefore signal-specific. Under the standard deviation model, P1 contains the lowest realized-volatility stocks within each sector. Under the maximum drawdown model, P1 contains the stocks with the lowest recent drawdowns within each sector. Under the beta model, P1 contains the lowest market-beta stocks within each sector.

Signal	Formula summary	P1 interpretation	Ranking direction
STDEV	Annualized 60-day return standard deviation	Lowest realized-volatility stocks	Low to high
MDD	Largest 60-day peak-to-trough loss	Lowest recent drawdown stocks	Low to high
BETA	60-day covariance with VNINDEX divided by market variance	Lowest market-sensitivity stocks	Low to high

4. Portfolio Construction

4.1 Monthly Formation Dates

The model is rebalanced monthly. Formation dates are defined as the last available trading date of each calendar month in the stock price matrix. At each formation date, the model observes the trailing risk signal, constructs portfolios, assigns weights, and holds each portfolio until the next monthly formation date.

The first raw data date does not necessarily equal the first backtest date. The strategy requires sufficient trailing price history before a valid signal can be computed. A 60-trading-day signal window means that the first feasible formation date occurs only after enough trading observations have accumulated and a monthly formation date is available.

4.2 Eligible Universe at Each Formation Date

For each signal and each monthly formation date, the model constructs an eligible stock universe. A stock is eligible if it satisfies all of the following conditions: it is inside the intended VN100 universe, it has a valid risk signal, it has a valid sector classification, it has a valid positive market capitalization observation, and it satisfies the minimum data requirement for the rolling signal. If the number of eligible stocks falls below the configured minimum universe size, the model skips that rebalance month for the affected signal.

4.3 Within-Sector Tertile Sort

Portfolio assignment is performed within each sector rather than across the full universe. This prevents the low-risk portfolio from becoming a concentrated bet on naturally defensive sectors and prevents the high-risk portfolio from becoming a concentrated bet on cyclical or high-beta sectors. At each formation date, stocks are grouped by sector and ranked by the relevant risk signal from lowest risk to highest risk.

Let $n_{s,t}$ be the number of stocks in that sector. The model defines:

$$n_{\text{top}} = \text{floor}\left(\frac{n_{s,t}}{3}\right), \quad n_{\text{bot}} = \text{floor}\left(\frac{n_{s,t}}{3}\right)$$

The n_{top} lowest-risk stocks are assigned to P1, the n_{bot} highest-risk stocks are assigned to P3, and the remaining stocks are assigned to P2. If a sector has fewer than the minimum sector size, all stocks in that sector are assigned to P2. This avoids unstable top-bottom splits in very small sector groups. Because P2 receives remainder stocks and small-sector stocks, P2 is normally larger than P1 and P3.

Portfolio	Risk position	Assignment rule
P1	Low-risk portfolio	Lowest floor($n/3$) stocks within each sector
P2	Middle-risk portfolio	Remaining middle stocks and small-sector stocks
P3	High-risk portfolio	Highest floor($n/3$) stocks within each sector

4.4 Sector-Replicated Market-Capitalization Weighting

After stocks are assigned to P1, P2, and P3, each portfolio is weighted using a sector-replicated market-capitalization method. The purpose is to preserve the sector composition of the eligible VN100 universe while still allowing the risk signal to determine stock selection within each sector.

Let U_t denote the eligible universe at formation date t . Let $P_{k,t}$ denote the set of stocks assigned to portfolio k , where k is P1, P2, or P3. Let $M_{i,t}$ denote the market capitalization of stock i at the formation date, and let $s(i)$ denote its sector. The universe-level sector weight is:

$$W_{s,t}^U = \frac{\sum_{i \in U_t, s(i)=s} M_{i,t}}{\sum_{i \in U_t} M_{i,t}} \quad (4.1)$$

For each portfolio k , only sectors represented inside that portfolio are retained. Let $R_t(k)$ denote the set of sectors represented in portfolio k . The target sector weight for represented sector s is:

$$W_{s,k,t}^T = \frac{W_{s,t}^U}{\sum_{s' \in R_t(k)} W_{s',t}^U}, \quad s \in R_t(k) \quad (4.2)$$

Within each represented sector, stocks are weighted by market capitalization relative to the total market capitalization of stocks in the same portfolio-sector group:

$$w_{i,k,t} = W_{s(i),k,t}^T \times \frac{M_{i,t}}{\sum_{j \in P_{k,t}, s(j)=s(i)} M_{j,t}} \quad (4.3)$$

Finally, weights are normalized to ensure that each portfolio-date pair sums to one:

$$\sum_{i \in P_{k,t}} w_{i,k,t} = 1 \quad (4.4)$$

This method means that each of P1, P2, and P3 is independently investable and approximately sector-neutral relative to the eligible VN100 universe. The market capitalization used for weighting is the most recent valid observation available on or before the formation date.

5. Trading Cost, Turnover, and Return Computation

5.1 Trading Cost Model

The final implementation applies explicit transaction costs in the market-capitalisation-weighted value, momentum, and combined modules. The configured trading cost assumptions are a 0.15% buy-side brokerage fee, a 0.15% sell-side brokerage fee, and a 0.10% sell-side tax. The implied round-trip cost is therefore 0.40%.

Cost component	Rate
Buy brokerage fee	0.15%
Sell brokerage fee	0.15%
Sell tax	0.10%
Total round-trip cost	0.40%

At each rebalance, the model distinguishes entering stocks, exiting stocks, and continuing holdings. Entering stocks are bought at their current target weights, and the buy fee is applied to those weights. Exiting stocks are sold at their drifted pre-rebalance weights, meaning the previous portfolio weights are first grown by the holding-period returns and then renormalised before applying the sell fee and sell tax. Continuing holdings incur cost only on the active rebalance trade: an increase in target weight receives the buy fee, while a decrease in target weight receives the sell fee plus sell tax.

$$\text{FeeDrag}_{k,t} = \text{BuyCost}_{k,t} + \text{SellCost}_{k,t} + \text{RebalanceCost}_{k,t} \quad (5.1)$$

$$\text{BuyCost}_{k,t} = \sum_{i \in B_{k,t}} w_{i,k,t} \times c^B \quad (5.2)$$

$$\text{SellCost}_{k,t} = \sum_{i \in S_{k,t}} w'_{i,k,t} \times (c^S + \tau^S) \quad (5.3)$$

$$\text{RebalanceCost}_{k,t} = \sum_{i \in H_{k,t}} [\max(w_{i,k,t} - w'_{i,k,t}, 0) c^B + \max(w'_{i,k,t} - w_{i,k,t}, 0) (c^S + \tau^S)] \quad (5.4)$$

where $B_{k,t}$ is the set of stocks entering portfolio k at formation date t , $S_{k,t}$ is the set of stocks exiting the portfolio, and $H_{k,t}$ is the set of stocks remaining in the portfolio. $w_{i,k,t}$ denotes the current target weight of stock i , while $w'_{i,k,t}$ denotes the drifted pre-rebalance weight based on the previous portfolio weights and realised returns since the prior rebalance. c^B is the buy-side brokerage fee, c^S is the sell-side brokerage fee, and τ^S is the sell-side transaction tax.

For the first rebalance of a portfolio, the model assumes the full portfolio is bought from cash, so turnover is set to 100% and the buy fee is applied to the full target portfolio weight.

5.2 Turnover

Turnover is computed as a weight-based measure rather than a stock-count measure in the market-capitalisation-weighted implementation. At each rebalance, the entering weight is the sum of target weights for newly added stocks, the exiting weight is the sum of drifted pre-rebalance weights for removed stocks, and the holding adjustment is the sum of absolute differences between target weights and drifted weights for stocks that remain in the portfolio.

$$\text{Turnover}_{k,t} = \frac{1}{2} \left(\sum_{i \in B_{k,t}} w_{i,k,t} + \sum_{i \in S_{k,t}} w'_{i,k,t} + \sum_{i \in H_{k,t}} |w_{i,k,t} - w'_{i,k,t}| \right) \quad (5.5)$$

The division by two prevents double-counting the buy and sell sides of the same reallocation. This measure better reflects actual traded portfolio capital than a simple count of changed tickers.

5.3 Gross and Net Return Computation

Gross return is computed as the market-capitalisation-weighted average of constituent stock returns over the relevant holding period. Stocks with missing holding-

period returns are excluded from the return calculation, and the remaining weights are effectively renormalised by dividing by the sum of weights with valid returns.

$$R_{k,t}^{gross} = \frac{\sum_{i \in \mathcal{V}_{k,t}} w_{i,k,t} * r_{i,t}}{\sum_{i \in \mathcal{V}_{k,t}} w_{i,k,t}} \quad (5.6)$$

$$R_{k,t}^{net} = R_{k,t}^{gross} - \text{FeeDrag}_{k,t} \quad (5.7)$$

where $R_{k,t}^{gross}$ is the gross return of portfolio k over holding period t , $R_{k,t}^{net}$ is the corresponding return after transaction costs, $w_{i,k,t}$ is the target portfolio weight of stock i , $r_{i,t}$ is the realised simple return of stock i over the holding period, and $\mathcal{V}_{k,t}$ denotes the subset of portfolio stocks with valid return observations. The denominator renormalises the portfolio weights when some constituent returns are missing.

6. Implementation Assumptions and Limitations

First, the framework is long-only. It does not implement short portfolios or long-short spread portfolios, even though the academic factor literature often evaluates value and momentum using long-short returns.

Second, the model depends on the input VN100 universe. If the same current VN100 list is used for the entire historical sample, the backtest may suffer from survivorship bias. A stricter historical backtest should use the actual VN100 membership at each formation date.

Finally, transaction costs are represented by fixed brokerage and tax rates. The model does not estimate bid-ask spreads, market impact, liquidity constraints, or execution slippage. The fee model is therefore a simplified implementation-cost approximation, consistent with the practical focus on turnover and fee drag emphasised by Frazzini, Israel and Moskowitz (2012).

7. Results and discussion

Table 1: Descriptive statistics of Net Portfolio Returns

	Beta			Maximum Drawdown			Standard Deviation		
	P1	P2	P3	P1	P2	P3	P1	P2	P3
Mean	13.02%	11.14%	17.27%	8.78%	12.85%	16.79%	7.71%	13.63%	21.17%
<i>(t-stat)</i>	(1.72)	(1.14)	(1.40)	(1.06)	(1.38)	(1.49)	(1.03)	(1.47)	(1.71)
Stdev	17.18%	22.15%	27.98%	18.98%	21.40%	25.77%	17.13%	21.30%	28.32%
Sharpe	0.76	0.50	0.62	0.46	0.60	0.65	0.45	0.64	0.75
Alpha	4.78%	-0.66%	2.28%	-1.19%	1.26%	2.91%	-1.12%	1.82%	5.76%
<i>(t-stat)</i>	(1.11)	(-0.17)	(0.48)	(-0.30)	(0.32)	(0.60)	(-0.29)	(0.54)	(1.14)
N	62	62	62	63	63	63	63	63	63

Note: The table reports descriptive statistics for net portfolio returns after transaction fee. Mean, standard deviation, and alpha are annualised using arithmetic annualisation. Alpha is estimated as the intercept from a single-factor regression on VN-INDEX. N denotes the number of return observations.

Table 1 reports the descriptive statistics of monthly net portfolio returns after transaction fees for the three volatility-based indicators: beta, maximum drawdown, and standard deviation. Across all specifications, the results show that the volatility signal does not generate a simple low-risk premium in raw return terms. In most cases, P3, which contains the highest-risk stocks, earns the highest annualised mean return. However, the risk-adjusted results are more mixed. Low-risk portfolios generally have lower return volatility, while high-risk portfolios earn higher raw returns but also carry substantially higher standard deviation. Therefore, the interpretation depends on whether the strategy objective is to maximize raw return, risk-adjusted return, or benchmark-adjusted alpha.

For the beta-sorted portfolios, the pattern is not monotonic in raw returns. P1, the lowest-beta portfolio, earns an annualised mean return of 13.02%, while P2 earns 11.14% and P3 earns 17.27%. Although P3 produces the highest average return, it also has the highest return volatility at 27.98%, compared with 17.18% for P1. As a result, P1 has the strongest Sharpe ratio among the beta portfolios at 0.76, compared with 0.50 for P2 and 0.62 for P3. This suggests that the beta signal is more useful for improving risk-adjusted performance than for identifying the portfolio with the highest raw return. The lowest-beta

portfolio gives up some upside relative to P3, but it achieves a more efficient return per unit of risk.

The alpha results reinforce this interpretation. The beta P1 portfolio generates an annualised alpha of 4.78% relative to VNINDEX, while P2 has a slightly negative alpha of -0.66% and P3 has a positive alpha of 2.28%. This indicates that, after controlling for market exposure, the low-beta portfolio delivers the strongest benchmark-adjusted performance among the beta-sorted portfolios. However, the alpha t-statistic for P1 is only 1.11, meaning that the alpha is not statistically strong under conventional significance levels. Therefore, the beta signal provides economically meaningful evidence of better risk-adjusted performance, but the statistical evidence remains limited.

The maximum-drawdown portfolios show a clearer raw-return pattern. Annualised mean return increases from 8.78% for P1 to 12.85% for P2 and 16.79% for P3. This means that stocks with larger recent drawdowns subsequently earn higher average returns in this sample. However, this higher return also comes with higher risk. The standard deviation increases from 18.98% for P1 to 21.40% for P2 and 25.77% for P3. The Sharpe ratio also rises from 0.46 for P1 to 0.60 for P2 and 0.65 for P3, suggesting that the higher-drawdown portfolio is not only earning a higher raw return but also achieving better risk-adjusted performance. This result is inconsistent with a low-drawdown defensive premium. Instead, it suggests that recent drawdown may behave more like a reversal or compensation-for-risk signal in the VN100 universe.

The alpha pattern for the maximum-drawdown portfolios also supports the same conclusion. P1 has a negative alpha of -1.19%, while P2 and P3 generate positive alphas of 1.26% and 2.91%, respectively. Although the alpha t-statistics are weak, with values of -0.30, 0.32, and 0.60, the direction of the results indicates that low-drawdown stocks do not outperform after controlling for VNINDEX exposure. The evidence therefore does not support using maximum drawdown as a defensive low-risk stock-selection signal. Under this specification, the highest-drawdown portfolio performs better than the lowest-drawdown portfolio, both in mean return and Sharpe ratio.

The standard-deviation portfolios provide the strongest evidence against a simple low-volatility return premium. Mean return rises monotonically across the three portfolios, from 7.71% for P1 to 13.63% for P2 and 21.17% for P3. At the same time, volatility also rises from 17.13% to 21.30% and 28.32%. This confirms that the standard-deviation sort successfully separates low-risk and high-risk stocks. However, the return pattern moves in the opposite direction from the low-volatility anomaly. The high-volatility P3 portfolio earns the highest average return and the highest Sharpe ratio at 0.75, compared with 0.45 for P1 and 0.64 for P2. Therefore, within this VN100 sample, high realized volatility appears to be rewarded rather than penalized.

The alpha results are consistent with the mean-return results. The standard-deviation P1 portfolio has a negative annualised alpha of -1.12%, while P2 and P3 produce positive alphas of 1.82% and 5.76%, respectively. The alpha t-statistic for P3 is 1.14, which is not conventionally significant, but it is still stronger than the corresponding statistics for P1 and P2. This suggests that the highest-volatility portfolio delivers the strongest benchmark-adjusted performance under the standard-deviation signal. Nevertheless, the lack of statistical significance means that the result should be interpreted as suggestive rather than conclusive.

Overall, Table 1 shows that the three volatility indicators behave differently. Beta produces the most favorable result for a defensive strategy because P1 achieves the highest Sharpe ratio and the strongest positive alpha among the beta portfolios. By contrast, maximum drawdown and standard deviation show stronger performance in P3, indicating that stocks with higher recent downside risk or higher realized volatility deliver better subsequent returns in the tested period. This finding suggests that the VN100 market does not exhibit a broad low-volatility anomaly under the current methodology. Instead, the evidence is more consistent with a positive risk-return trade-off for standard deviation and maximum drawdown, while beta provides a more useful defensive signal.

The statistical strength of the results remains modest. Most mean-return t-statistics are between approximately 1.0 and 1.7, and most alpha t-statistics are below 1.2. This

means that the observed differences are economically meaningful but not statistically decisive. Therefore, the results should not be interpreted as definitive evidence that one volatility indicator consistently predicts returns. A more cautious interpretation is that standard deviation and maximum drawdown identify portfolios with clearly different risk profiles, and that higher-risk portfolios earn higher average returns during the sample period. Beta, meanwhile, offers the strongest evidence of improved risk-adjusted performance for the low-risk portfolio.

Table 2: Portfolio Turnover and Fee-Adjusted Return

Indicators	Portfolios	Annualized gross return	Average turnover	Annualized fee drag	Annualized net return
Beta	P1	14.44%	30.44%	1.41%	13.02%
	P2	13.30%	45.69%	2.15%	11.14%
	P3	18.96%	35.68%	1.67%	17.27%
Maximum Drawdown	P1	10.78%	42.22%	1.98%	8.78%
	P2	15.44%	54.46%	2.57%	12.85%
	P3	19.08%	48.23%	2.27%	16.79%
Standard Deviation	P1	8.99%	27.83%	1.29%	7.71%
	P2	15.75%	44.84%	2.11%	13.63%
	P3	22.79%	34.56%	1.61%	21.17%

Table 3 reports portfolio turnover, transaction-cost drag, and fee-adjusted returns for the three volatility indicators. The results show that transaction costs reduce performance across all portfolios, but they do not fully explain the ranking of returns. Since the strategy is rebalanced monthly rather than weekly, the difference between gross and net returns is noticeable but not large enough to overturn the main performance patterns. The return differences across P1, P2, and P3 are mainly driven by the volatility signals themselves rather than by transaction costs.

For the beta-sorted portfolios, fee drag is moderate across all three groups. P1 has average turnover of 30.44% and annualised fee drag of 1.41%, reducing gross return from 14.44% to a net return of 13.02%. P2 has the highest turnover among the beta portfolios at 45.69%, which leads to the highest annualised fee drag of 2.15%. As a result, its gross return of 13.30% falls to a net return of 11.14%. P3 has average turnover of 35.68% and fee drag of 1.67%, leaving a net return of 17.27% from a gross return of 18.96%. These results show that transaction costs affect all beta portfolios, but the cost impact is not severe. P3 remains the strongest beta portfolio in raw return terms after fees, while P1 remains attractive from a defensive perspective because it delivers a relatively high net return with lower turnover and lower risk.

The maximum drawdown portfolios show higher turnover than the beta portfolios. P1 has average turnover of 42.22%, P2 has 54.46%, and P3 has 48.23%. This produces annualised fee drag of 1.98%, 2.57%, and 2.27%, respectively. P2 has the highest transaction-cost burden in the full table, reflecting greater constituent instability in the middle-drawdown group. Despite this, the return ranking remains clear after fees. P1 earns a gross return of 10.78% and a net return of 8.78%. P2 earns a gross return of 15.44% and a net return of 12.85%. P3 earns a gross return of 19.08% and a net return of 16.79%. This indicates that the high-drawdown portfolio's stronger performance is not simply an artifact of ignoring transaction costs. Even after a relatively high fee drag, P3 still produces the highest net return among the maximum drawdown portfolios.

The standard deviation portfolios show the clearest return spread after transaction costs. P1 has the lowest average turnover in the table at 27.83%, resulting in annualised fee drag of only 1.29%. However, its gross return is also the lowest at 8.99%, leaving a net return of 7.71%. P2 has higher turnover of 44.84% and annualised fee drag of 2.11%, reducing gross return from 15.75% to a net return of 13.63%. P3 has average turnover of 34.56% and annualised fee drag of 1.61%, but it earns the highest gross return of 22.79% and the highest net return of 21.17%. This result is important because P3 does not outperform by taking excessive turnover. Its turnover is lower than P2 and only moderately higher than P1. Therefore, the strong net performance of the high-standard-deviation

portfolio is mainly explained by its higher gross return rather than by lower implementation cost.

Across the three indicators, the fee-adjusted results confirm that monthly rebalancing is implementable under the assumed transaction-cost model. The highest annualised fee drag is 2.57% for the maximum drawdown P2 portfolio, while most portfolios have fee drag between 1.3% and 2.3%. These values are economically relevant but not large enough to eliminate gross profitability. All nine portfolios retain positive net returns after transaction costs. This suggests that the monthly volatility strategy is less vulnerable to fee erosion than a weekly strategy with frequent reshuffling.

The results also show that transaction costs do not change the main conclusion from the descriptive return statistics. For beta, P3 has the highest net return, but P1 remains comparatively efficient because it has lower turnover, lower fee drag, and a strong net return. For maximum drawdown, P3 continues to dominate P1 and P2 after fees, suggesting that stocks with larger recent drawdowns earned higher subsequent returns in this sample. For standard deviation, P3 remains the best-performing portfolio by a wide margin, with a net return of 21.17% compared with 7.71% for P1. This reinforces the earlier finding that the standard deviation signal does not produce a low-volatility return premium. Instead, the highest-volatility group earns the strongest fee-adjusted return.

From an implementation perspective, Table 3 suggests that the volatility model is not mainly constrained by transaction costs. The more important issue is the direction of the signal. If the strategy objective is to build a defensive portfolio, the beta P1 portfolio appears more suitable because it combines moderate return, lower turnover, and lower risk. If the objective is to maximize net return, the standard deviation P3 portfolio is the strongest candidate, followed by beta P3 and maximum drawdown P3. However, this comes with higher risk exposure, as shown in the descriptive statistics. Therefore, the practical choice depends on whether the investor prioritizes absolute return or risk-adjusted defensive performance.

Overall, the study should be interpreted as an exploratory backtest of volatility-based portfolio construction within VN100. Its main limitations are the restricted universe, possible survivorship bias, limited monthly sample size, simplified transaction-cost assumptions, reliance on a single 60-day volatility window, and benchmark adjustment using only VNINDEX. The sector-neutral and market-cap-weighted construction improves investability but may dilute the raw volatility signal. The results are therefore useful for evaluating whether volatility indicators separate future portfolio performance, but they do not by themselves prove that the strategy is robust, statistically persistent, or directly implementable in live trading without further validation.

8. References

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